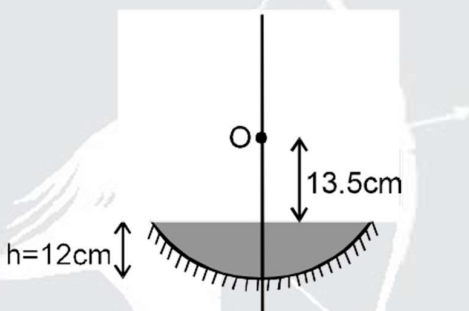
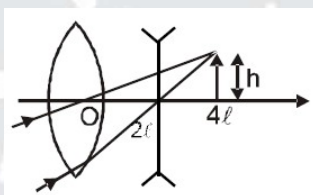


1. A concave mirror of radius of curvature 40 cm is filled with water ($n = 4/3$) upto a height of 12 cm. A point object O is kept on the principal axis of the mirror at height 13.5 cm from the water surface. The final image formed after refraction at water surface, reflection at mirror and again refraction at water surface in succession is situated at.

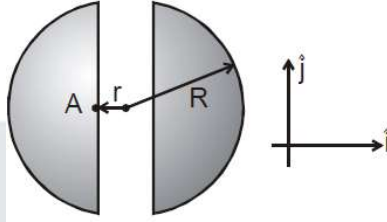


- (A) 36 cm above the water surface (B) 24 cm above the water surface
(C) 20 cm above the water surface (D) 12 cm above the water surface
2. A thin converging lens L_1 forms a real image of an object located far away from the lens as shown in the figure. The image is located a distance 4ℓ and has height 'h'. A diverging lens of focal length ℓ is placed 2ℓ from lens L_1 . Another converging lens of focal length 2ℓ is placed 3ℓ from lens L_1 . The height of final image thus formed is (Both diverging and converging lenses are placed at right side of L_1)-



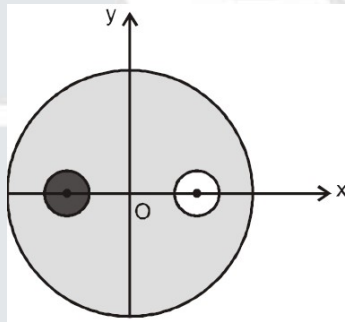
- (A) h (B) $\frac{h}{2}$ (C) 4 h (D) 2 h

3. A cylindrical portion of radius 'r' is removed from a solid sphere of radius R and uniform volume charge density ρ in such a way that the axis of the hollow cylinder coincides with one of the diameters of the sphere. ('r' is negligible compared to R). Then the electric field intensity at point A is



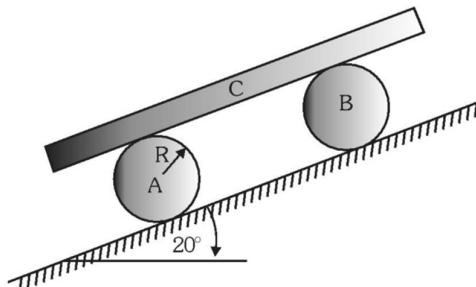
- (A) $\frac{\rho r}{3\epsilon_0} \hat{i}$ (B) $-\frac{\rho r}{3\epsilon_0} \hat{i}$ (C) $\frac{\rho r}{6\epsilon_0} \hat{i}$ (D) $-\frac{\rho r}{6\epsilon_0} \hat{i}$

4. Consider a solid sphere of density ρ and radius $4R$. Centre of the sphere is at origin. Two spherical cavities centered at $(2R, 0)$ and $(-2R, 0)$ are created in sphere. Radii of both cavities is R . In left cavity material of density 2ρ is filled while second cavity is kept empty. What is gravitational field at origin.



- (A) $\frac{G\rho\pi R}{3}$ (B) $\frac{2G\rho\pi R}{3}$ (C) $\frac{4G\rho\pi R}{3}$ (D) $\frac{3G\rho\pi R}{2}$

5. Cylinders A and B each have a mass of 25 kg and diameter of 0.3 m. Block C, riding on A and B, has a mass of 100 kg. If the system is released from rest at the configuration shown, what is the speed of C after the cylinders have made half a revolution? Slipping is absent everywhere. (Given $\sin 20^\circ = 0.34$; $g = 10 \text{ m/s}^2$)

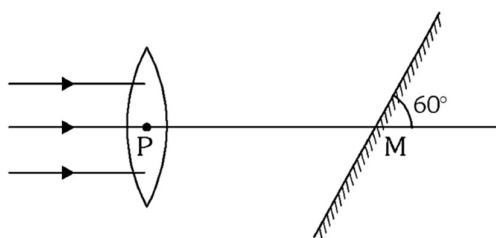


(A) 1.3 m/s (B) 5.6 m/s (C) 10.1 m/s (D) 15.5 m/s

6. A cyclist is riding a bicycle at a speed of 5 m/s on a horizontal road. The distance between the axles is 2 m, and the mass center of the cyclist and the bicycle is located 1 m behind the front axle and 2 m above the ground. If the cyclist applies the brakes only on the front wheel, determine the shortest distance in which he can stop without being thrown over the front wheel. (Take $g = 10 \text{ m/s}^2$)

(A) 0.5 (B) 1.5 m (C) 2.5 m (D) 3.5 m

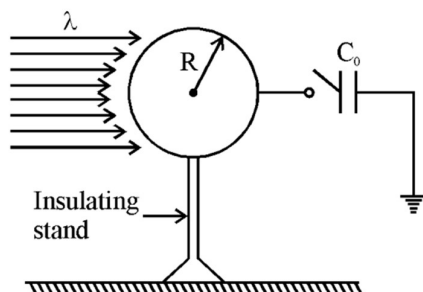
7. A horizontal parallel beam of light passes through a vertical convex lens of focal length f . The optical centre of the lens is P . A small plane mirror is placed at point M inclined at 60° to the axis of the lens. Distance $PM = f/2$. The mirror reflects the light passing through the lens and forms an image at point I . Find distance PO .



(A) f (B) $\frac{f}{2}$ (C) $\frac{f}{4}$ (D) $\frac{f}{6}$

8. An insulated conducting spherical shell is exposed to a radiation of wavelength λ . The threshold wavelength of the conducting shell is, $\lambda_0 (> \lambda)$. Since energy of incident photon is greater than threshold energy, the emission of photoelectrons will occur and due to it sphere gets charged and its potential becomes V_0 and at this moment emission of photoelectrons stops. A parallel plate capacitor of capacitance C_0 is connected to shell through a switch.

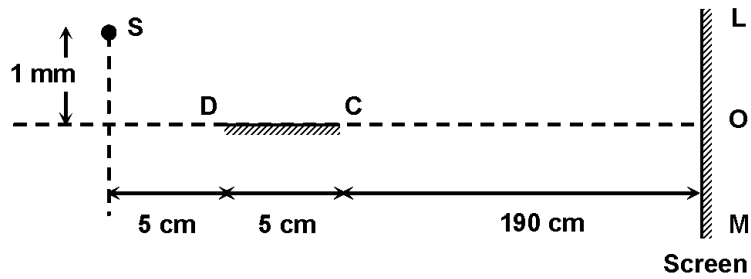
(Given : $\frac{hc}{e} = 12 \times 10^{-7} \text{ V-m}$, $\lambda = 4000 \text{ \AA}$, $\lambda_0 = 6000 \text{ \AA}$, $C_0 = 2 \mu\text{F}$, $R = 18 \text{ cm}$)



By closing the switch a capacitor is connected to sphere as shown in figure. The charge on the capacitor after long time is (in μC) :

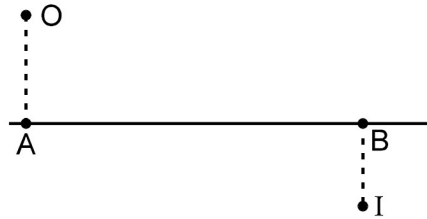
(A) 2 (B) 0.5 (C) 1 (D) 10^{-5}

9. A monochromatic source of light of wavelength $5000 \text{ \AA}$ is placed in situation as shown in figure. Considering interference of direct and reflected rays by a plane mirror DC.



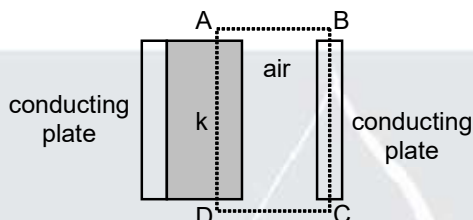
- (A) The length (along OL) of the region where the fringes will be visible is 2 cm
 (B) The fringe width is 0.05 cm
 (C) The number of fringes is 20
 (D) The number of fringes is 40
10. In a hydrogen like ion, nucleus has a positive charge Ze Bohr's quantization rule is, the angular momentum of an electron about the nucleus $\ell = \frac{nh}{2\pi}$, where n is a positive integer.
- (A) If electron goes from ground state to 1st excited state then change in energy of the hydrogen like ion is $\frac{3}{32} \frac{m_e z e^4}{\epsilon_0^2 h^2}$
- (B) Change in K.E. of electron when it goes from ground state to 1st excited state is $\frac{-3}{32} \frac{m_e z e^4}{\epsilon_0^2 h^2}$
- (C) If electron is in ground state the magnetic field produced at the site of nucleus due to circular motion of the electron is $\frac{\mu_0 \pi Z^3}{8} \frac{e^7 m^2}{\epsilon_0^3 h^5}$
- (D) If electron goes from ground state to 1st excited state then change in energy of the hydrogen like ion is $\frac{3}{16} \frac{m_e z e^4}{\epsilon_0^2 h^2}$

11. A luminous point object is placed at O, whose image is formed at I as shown in figure. Line AB is the optical axis. Which of the following statement is/are correct ?



- (A) If a lens is used to obtain the image, then it must be a converging lens and its optical centre will be the intersection point of line AB and OI.
- (B) If a lens is used to obtain the image, then it must be a diverging lens and its optical centre will be the intersection point of line AB and OI.
- (C) If a mirror is used to obtain the image then the mirror must be concave and object and image subtend equal angles at the pole of the mirror.
- (D) I is a real Image.
12. A glass capillary tube of 0.11 m length and 2×10^{-5} m internal radius has its one end sealed. The tube is dipped vertically in a liquid of surface tension 5.06×10^{-2} N/m. If the liquid level inside and outside the tube be the same, then (atmospheric pressure $p_a = 1 \times 10^5$ N/m²) (liquid is water)
- (A) Pressure of air inside the tube over and above the atmosphere is approximately 5 kN/m².
- (B) Length of the tube inside the liquid is 5.23×10^{-3} m approximately.
- (C) If the capillary tube is broken, water will overflow.
- (D) If the capillary tube is broken, water level will fall.
13. An earth satellite near earth's surface and a moon satellite near moon's surface take equal time per revolution. Which of the following can be deduced from this information ?
Mass of earth (M_e) > Mass of moon (M_m) ; Radius of earth (R_e) > Radius of moon (R_m)
- (A) Both earth and moon have equal density
- (B) Earth has higher density than moon
- (C) Moon has higher density than earth
- (D) $\frac{M_m}{M_e} = \frac{r_m^3}{r_e^3}$

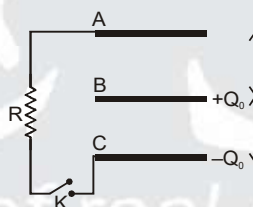
14. The figure shows two thin large identical parallel conducting plates placed in air with a slab of dielectric constant k placed between them. The width of the small gap between the conducting plates is d while the thickness of the dielectric slab is t . The conducting plates are given equal and opposite charges due to which the strength of the electric field in air region is E as shown. The area of plates perpendicular to the plane of figure is A and consider it to be large. Neglect the edge effects. (The portion ABCD has the same dimensions perpendicular to the plane of the figure as that of plate or dielectric medium) Mark the correct options :



- (A) The net charge in the portion ABCD (volume) is zero.
 (B) The magnitude of net charge in the portion ABCD is $\frac{\epsilon_0 EA}{k}$
 (C) The electrostatics energy stored in the dielectric medium is $\frac{\epsilon_0 E^2 At}{2k}$
 (D) The electrostatic energy stored in the dielectric medium is $\frac{\epsilon_0 E^2 At}{2k^2}$

Paragraph for Question Nos. 15 to 16

Three identical metal plates of area S are at equal distances b as shown. Initially metal plate A is uncharged, while metal plates B and C have respective charges $+Q_0$ and $-Q_0$ initially as shown. Metal plates A and C are connected by switch K through a resistor of resistance R . The key K is closed at time $t = 0$.



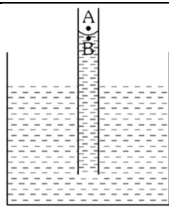
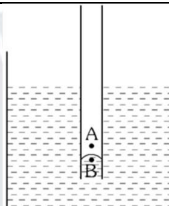
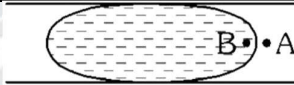
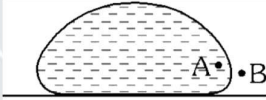
15. Then the magnitude of current in amperes through the resistor at any later time t is :

- (A) $\frac{Q_0}{RS\epsilon_0} b e^{\frac{-bt}{RS\epsilon_0}}$ (B) $\frac{Q_0}{RS\epsilon_0} b e^{\frac{-2bt}{RS\epsilon_0}}$ (C) $\frac{Q_0}{2RS\epsilon_0} b e^{\frac{-2bt}{RS\epsilon_0}}$ (D) $\frac{Q_0}{2RS\epsilon_0} b e^{\frac{-bt}{RS\epsilon_0}}$

16. The total heat produced dissipated by resistor of resistance R is :

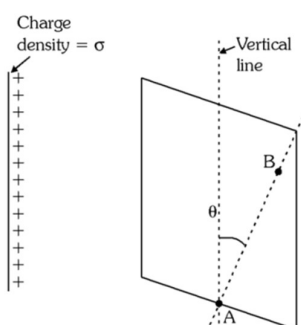
- (A) $\frac{Q_0^2}{4S} \frac{b}{\epsilon_0}$ (B) $\frac{Q_0^2}{8S} \frac{b}{\epsilon_0}$ (C) $\frac{Q_0^2}{2S} \frac{b}{\epsilon_0}$ (D) $\frac{3}{4S} \frac{Q_0^2}{\epsilon_0} b$

17. Capillary rise and shape of droplets on a plate due to surface tension are shown in column-II. Match the following:

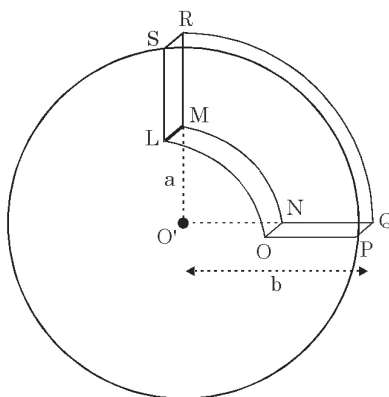
	List-I		List-II
(P)	Adhesive forces is greater than cohesive force	(1)	
(Q)	Cohesive forces is greater than adhesive forces	(2)	
(R)	Pressure at A > pressure at B	(3)	
(S)	Pressure at B > Pressure at A	(4)	

- (A) P → 2; Q → 3,4; R → 1,4; S → 1,3 (B) P → 3; Q → 2,3; R → 1,4; S → 1,2
 (C) P → 1; Q → 2,3,4; R → 1,4; S → 2,3 (D) P → 4; Q → 1,4; R → 1,2; S → 2,3
18. A satellite with mass 2000 kg and angular momentum magnitude $2 \times 10^{12} \text{ kg.m}^2/\text{s}$ is moving in an elliptical orbit around a planet. The rate at which area is being swept out by the satellite around the planet, is equal to $\alpha \times 10^8 \text{ m}^2/\text{s}$. Find α
19. A very large sheet is kept parallel to a vertical wall. A point charge is fixed at point A and another point charge 1C and mass 0.1 kg is at point B on the wall. For the particle at B to stay in equilibrium the maximum value of θ is $\sin^{-1}\left(\frac{1}{n}\right)$. Find n. Coefficient of friction between wall

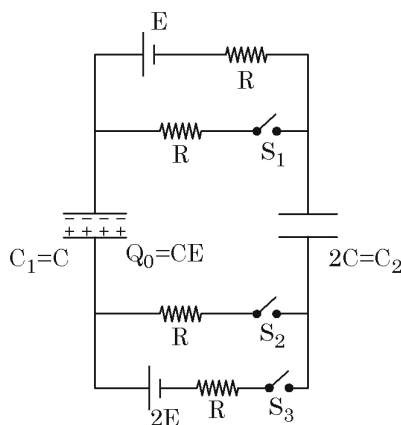
and charge particle B is $\mu = \frac{1}{9} \cdot \left(\frac{\sigma}{2\epsilon_0} = 1 \text{ N/C}, g = 10 \text{ m/s}^2 \right)$



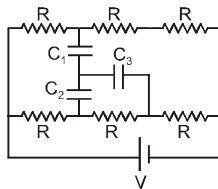
20. A conducting medium is shaped in a form of a quarter of an annulus of radii b and a ($b > a$) and thickness t . Specific resistance of the medium is ρ . Resistance between faces LMNO and PQRS is $\frac{k\rho}{\pi t} \ln \frac{b}{a}$. Find the value of k .



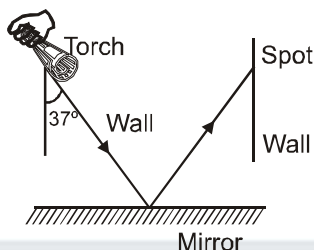
21. In given circuit diagram initial charge on parallel plates of capacitor C_1 is shown in figure. Now all the switches are closed simultaneously. Find the ratio of electrostatic force of attraction between plates of capacitor C_1 before closing the switches to steady state reached after closing the switches.



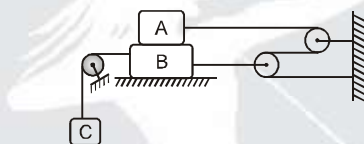
22. In the shown circuit, all three capacitor are identical and have capacitance $C \mu F$ each. Each resistor has resistance of $R \Omega$. An ideal cell of emf $V = 108$ volt is connected as shown. Then the magnitude of potential difference across capacitor C_3 in steady state is :



23. As shown in the figure a torch is used to send a beam of light on a mirror which forms a spot on a vertical wall. If the mirror is moved 4cm upwards and the wall 6 cm towards left, keeping the direction of beam fixed, the vertical shift of spot on the wall is x cm, then value of x is



24. Block A of weight 50 N and block B of weight 70 N are connected by rope pulley system as shown. The largest weight C that can be suspended without moving block A and B is W. The coefficient of friction for all plane surfaces of contact is 0.3. If the pulleys are ideal, then the value of W (in N) is .



25. A particle moves in the x-y plane with constant acceleration 'a' directed along the negative y-axis. Relation between x and y co-ordinate of the path followed by the particle is $y = px - qx^2$ where p and q are positive constants. Velocity of the particle at the origin of co-ordinate system is $\sqrt{\frac{a(\alpha + p^2)}{\beta q}}$. Here α and β are integers. Find $\alpha + \beta$.

ANSWER KEY

1. (A)	2. (D)	3. (C)	4. (B)	5. (A)
6. (C)	7. (B)	8. (A)	9. (ABD)	
10. (ABC)	11. (ACD)	12. (AB)	13. (AD)	14. (BC)
15. (B)	16. (A)	17. (C)	18. 5	19. 9
20. 2	21. 2.25	22. 24	23. 0	24. 81
25. +3				